STAT6611

- Lecture 15 (T, Oct 13)
- solutions posted to Homework 2
- Homework 3 assigned due F, Oct 16 by midnight
- Take home midterm to be given on R, Oct 15 & due F, Oct 23 (one week)
- We'll cancel class on R, Oct 15
- I'll fix our canvas front-page

Theorem 3.7. *Suppose that* $\underline{x} \sim N(\underline{\mu}, \Sigma)$. *Partition* $\underline{x}, \underline{\mu}$, *and* Σ *conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \vdots \\ \underline{\mu}_m \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \cdots & \Sigma_{1m} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \cdots & \Sigma_{mm} \end{bmatrix},$$

where $m \geq 2$. *Then,* $\underline{x}_1, \ldots, \underline{x}_m$ *are mutually independent if and only if* $\Sigma_{ij} = 0$ *for* $i = 1, \ldots, m$, $j = 1, \ldots, m$, *and* $i \neq j$.

↑ $\text{cov}(\underline{x}_i, \underline{x}_j) = 0$

Proof. For the "if" part ($\Longrightarrow$), suppose that $\Sigma_{ij} = 0$ for $i = 1, \ldots, m$, $j = 1, \ldots, m$, and $i \neq j$. Let $\underline{t} = \begin{bmatrix} \underline{t}_1 \\ \vdots \\ \underline{t}_m \end{bmatrix}$, where $\underline{t}_i$ and $\underline{x}_i$ have the same dimensions for $i = 1, \ldots, m$. Then,

joint mgf

$$M_{\underline{x}}(\underline{t}) = \exp\left[\underline{t}'\underline{\mu} + \left(\frac{1}{2}\right)\underline{t}'\Sigma\underline{t}\right] = \exp\left[\sum_{i=1}^{m}\underline{t}_i'\underline{\mu}_i + \left(\frac{1}{2}\right)\sum_{i=1}^{m}\sum_{j=1}^{m}\underline{t}_i'\Sigma_{ij}\underline{t}_j\right]$$

by def

$\Sigma_{ij} = 0$ $i \neq j$

$$= \exp\left[\sum_{i=1}^{m}\underline{t}_i'\underline{\mu}_i + \left(\frac{1}{2}\right)\sum_{i=1}^{m}\underline{t}_i'\Sigma_{ii}\underline{t}_i\right] = \prod_{i=1}^{m}\exp\left[\underline{t}_i'\underline{\mu}_i + \left(\frac{1}{2}\right)\underline{t}_i'\Sigma_{ii}\underline{t}_i\right] = \prod_{i=1}^{m} M_{\underline{x}_i}(\underline{t}_i),$$

since $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$ by Theorem 3.6.

$M_{\underline{x}_i}(\underline{t}_i)$ since $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$

Since $M_{\underline{x}}(\underline{t})$ is a product,
the $\underline{x}_1, \ldots \underline{x}_m$ are independent

$\Leftarrow$ since $\underline{x}_i, \underline{x}_j$ (have variances) are independent
$\Sigma_{ij} = \text{cov}(\underline{x}_i, \underline{x}_j) = 0$ holds
ie $\text{cov}(\underline{x}_i, \underline{x}_j) = E[(\underline{x}_i - \underline{\mu}_i)(\underline{x}_j - \underline{\mu}_j)']$
ind $= E(\underline{x}_i - \underline{\mu}_i) \cdot E(\underline{x}_j - \underline{\mu}_j)' = 0$
$0 \quad 0$

Corollary. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. Let B_1 and B_2 be $q \times p$ and $r \times p$ matrices of constants. Let $\underline{b}_1$ and $\underline{b}_2$ be $q \times 1$ and $r \times 1$ vectors of constants. Then, the random vectors

$$\underline{y}_1 = B_1\underline{x} + \underline{b}_1 \quad \text{and} \quad \underline{y}_2 = B_2\underline{x} + \underline{b}_2$$

are independent if and only if $B_1 \Sigma B_2' = 0$.

Proof. By Theorem 3.5,

$$\underline{y} = \begin{bmatrix} \underline{y}_1 \\ \underline{y}_2 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} \underline{x} + \begin{bmatrix} \underline{b}_1 \\ \underline{b}_2 \end{bmatrix}$$

has a $(q + r)$-variate normal distribution. Then, by Theorem 3.7, $\underline{y}_1$ and $\underline{y}_2$ are independent if and only if $\Sigma_{12} = \text{cov}(\underline{y}_1, \underline{y}_2) = B_1 \Sigma B_2' = 0$. □

Theorem 3.8 is often used for Theorem 3.9 (conditional distributions).

Theorem 3.8. *Let* $M = \begin{bmatrix} A & C \\ D & B \end{bmatrix}$ *be a partitioned matrix such that* M*,* A*, and* B *are square matrices. If* B *is nonsingular, then* $|M| = |B| \cdot |A - CB^{-1}D|$*. Similarly, if* A *is nonsingular, then* $|M| = |A| \cdot |B - DA^{-1}C|$*.*

Theorem 3.9. *Suppose that* $\underline{x} \sim N(\underline{\mu}, \Sigma)$ *and* Σ *is nonsingular. Partition* $\underline{x}, \underline{\mu}$*, and* Σ *conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \underline{x}_2 \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}.$$

Then, the conditional distribution of $\underline{x}_1$ *given* $\underline{x}_2$ *is normal with mean vector* $\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2)$ *and variance-covariance matrix* $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$*.*

Example 1. Let $\underline{z} = (z_1, \ldots, z_p)' \sim N(\underline{0}, I)$. Let $\underline{y} = (z_1 - \overline{z}, \ldots, z_p - \overline{z})'$, where $\overline{z} = \left(\sum_{i=1}^{p} z_i\right)/p$. Find the distribution of $\underline{y}$.

Let $\underline{1} = (1, \ldots, 1)'$ be a $p \times 1$ vector of 1's. Then, $\overline{z} = \underline{1}'\underline{z}/p$ and

$$\underline{y} = \underline{z} - \overline{z}\,\underline{1} = \underline{z} - \left(\underline{1}'\underline{z}/p\right)\underline{1} = \underline{z} - \underline{1}\left(\underline{1}'\underline{z}/p\right) = \underline{z} - \left(\underline{1}\,\underline{1}'/p\right)\underline{z} = \left(I - \underline{1}\,\underline{1}'/p\right)\underline{z}.$$

Note that

$$\left(\underline{1}\,\underline{1}'/p\right)\left(\underline{1}\,\underline{1}'/p\right) = \underline{1}\left(\underline{1}'\underline{1}/p\right)\underline{1}'/p = \underline{1}\,\underline{1}'/p.$$

$\underline{1}\,\underline{1}'/p$ and $I - \underline{1}\,\underline{1}'/p$ are symmetric/ idempotent matrices so that (corollary/Lemma 2.5)

$$\operatorname{rank}\left(I - \underline{1}\,\underline{1}'/p\right) = \operatorname{tr}\left(I - \underline{1}\,\underline{1}'/p\right) = \operatorname{tr}(I) - \operatorname{tr}\left(\underline{1}\,\underline{1}'/p\right) = p - \operatorname{rank}\left(\underline{1}\,\underline{1}'/p\right) = p - 1.$$

Then, by Theorem 3.5 (or the definition of the p-variate normal distribution), $\underline{y}$ has a p-variate normal distribution of rank $p-1$, and $\underline{y} \sim N(\underline{0}, I - \underline{1}\,\underline{1}'/p)$.

Remark. Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector and $\text{rank}(\Sigma) = k \geq 1$. We can retrieve a k-dimensional standard normal vector from $\underline{x}$ as follows.

Take Γ as a $k \times p$ matrix of rank k such that $\Sigma = \Gamma'\Gamma$ and define

$$\underline{y} = (\Gamma\Gamma')^{-1}\Gamma(\underline{x} - \underline{\mu}),$$

which has a k-variate standard normal distribution, because $E(\underline{y}) = \underline{0}$ and

$$\text{Var}(\underline{y}) = (\Gamma\Gamma')^{-1}\Gamma\Sigma\Gamma'(\Gamma\Gamma')^{-1} = (\Gamma\Gamma')^{-1}\Gamma\Gamma'\Gamma\Gamma'(\Gamma\Gamma')^{-1} = I_k.$$

So, $\underline{Y} \sim MVN(\underline{0}, I_k)$ ($\underline{Y}$ is k-dim standard normal)

Theorem 3.6. *Suppose that $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Partition $\underline{x}, \underline{\mu}$, and Σ conformally as*

$$\underline{x} = \begin{bmatrix} \underline{x}_1 \\ \vdots \\ \underline{x}_m \end{bmatrix}, \quad \underline{\mu} = \begin{bmatrix} \underline{\mu}_1 \\ \vdots \\ \underline{\mu}_m \end{bmatrix}, \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \cdots & \Sigma_{1m} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \cdots & \Sigma_{mm} \end{bmatrix},$$

where $m \geq 2$. Then, $\underline{x}_i \sim N(\underline{\mu}_i, \Sigma_{ii})$ for $i = 1, \ldots, m$.

proof: $\underline{X}_i$ is a linear combination of $\underline{X}$ so $\underline{X}_i$ is $N(\underline{\mu}_i, \Sigma_{ii})$ by theorem 3.5 ($E(\underline{X}_i)$, $\text{Var}(\underline{X}_i)$)

Example 2. Let $\underline{x} = (x_1, x_2, x_3, x_4)' \sim N(\underline{\mu}, \Sigma)$, where

$$\underline{\mu} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 3 & 0 & 2 \\ 1 & 0 & 5 & 0 \\ 0 & 2 & 0 & 3 \end{bmatrix}.$$

(a) Find the distribution of $y = x_1 + 2x_2 + 3x_3 + x_4 + 2$.

Let $\underline{a} = (1, 2, 3, 1)'$. Then, $y = \underline{a}'\underline{x} + 2$. By Theorem 3.5, $y \sim N(\underline{a}'\underline{\mu} + 2, \underline{a}'\Sigma\underline{a})$, or $y \sim N(7, 76)$ (using R for numerical calculations).

(b) Find the marginal distribution of $\underline{x}_0 = (x_1, x_3)'$.

Let $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. Then, $\underline{x}_0 = B\underline{x}$. By Theorem 3.5, $\underline{x}_0 \sim N(\underline{\mu}_0, \Sigma_0)$, where

$$\underline{\mu}_0 = B\underline{\mu} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \Sigma_0 = B\Sigma B' = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}.$$

(c) Find the conditional pdf of $\underline{x}_1 = (x_1, x_2)'$ given $\underline{x}_2 = (x_3, x_4)'$.

By Theorem 3.9, the conditional pdf of $\underline{x}_1$ given $\underline{x}_2$ is

$$n\left[\underline{x}_1 : \underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}\right],$$

where

$$\underline{\mu}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \ \underline{\mu}_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \ \Sigma_{11} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \ \Sigma_{12} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \ \Sigma_{21} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix},$$

$$\Sigma_{22} = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}, \quad \text{and thus} \quad \Sigma_{22}^{-1} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}.$$

Then,

$$\underline{\mu}_1+\Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2-\underline{\mu}_2) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}+\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}\begin{bmatrix} x_3-2 \\ x_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{5}(x_3+3) \\ \frac{2}{3}x_4-1 \end{bmatrix},$$

and

$$\Sigma_{11}-\Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}-\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{9}{5} & 0 \\ 0 & \frac{5}{3} \end{bmatrix}.$$

Thus, the conditional pdf of $\underline{x}_1$ given $\underline{x}_2$ is

$$n\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}:\begin{bmatrix} \frac{1}{5}(x_3+3) \\ \frac{2}{3}x_4-1 \end{bmatrix},\begin{bmatrix} \frac{9}{5} & 0 \\ 0 & \frac{5}{3} \end{bmatrix}\right).$$

Note that the marginal pdf of $\underline{x}_1$ is

$$n\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}:\begin{bmatrix} 1 \\ -1 \end{bmatrix},\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\right).$$

We have that

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}-\begin{bmatrix} \frac{9}{5} & 0 \\ 0 & \frac{5}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & 0 \\ 0 & \frac{4}{3} \end{bmatrix}$$

is a positive definite matrix. Thus, the conditional distribution of $\underline{x}_1$ given $\underline{x}_2$ has a "smaller" variance-covariance matrix than that of the marginal distribution of $\underline{x}_1$.

3.2 Moment Generating Function & Variance of a Quadratic Form

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is a $p \times 1$ vector. Take A to be a $p \times p$ symmetric matrix of constants. Then, for t such that $|t|$ is sufficiently small, it holds that

$$M_{\underline{x}' A \underline{x}}(t) = |I - 2tA\Sigma|^{-1/2} \exp\left[t\underline{\mu}' \left(I - 2tA\Sigma\right)^{-1} A\underline{\mu}\right].$$

Proof. For t with small sufficiently small $|t|$ and writing $\Sigma = \Gamma'\Gamma$ with $k \times p$ matrix Γ with $k = \text{rank}(\Sigma)$ and k-variate standard normal $\underline{z}$,

$$\begin{aligned}
M_{\underline{x}' A \underline{x}}(t) &= E \exp\left(t\underline{x}' A \underline{x}\right) = E \exp\left(t(\underline{\mu} + \Gamma' \underline{z})' A (\underline{\mu} + \Gamma' \underline{z})\right) \\
&= (2\pi)^{-k/2} \int_{\mathbb{R}^k} \exp\left[-\frac{1}{2}\underline{z}'(I - 2t\Gamma A \Gamma')\underline{z} + 2t\underline{z}'\Gamma A \underline{\mu} + t\underline{\mu}' A \underline{\mu}\right] d\underline{z} \\
&= |I - 2t\Gamma A \Gamma'|^{-1/2} \exp\left[2t^2 \underline{\mu}' A \Gamma' (I - 2t\Gamma A \Gamma')^{-1} \Gamma A \underline{\mu} + t\underline{\mu}' A \underline{\mu}\right] \\
&= |I - 2tA\Sigma|^{-1/2} \exp\left[t\underline{\mu}' (I - 2tA\Sigma)^{-1} A\underline{\mu}\right]
\end{aligned}$$

applying Theorem 3.6/Lemma 3.1 for integration and re-arranging by corollary/Theorem 3.10 with $S = -2tA\Gamma'$, $U = \Gamma$, and by Theorem 1.5 with $R = I$, $S = -2tA\Gamma'$, $T = I$, and $V = \Gamma$, so $(I + SU)^{-1} = I - S(I + US)^{-1}U$. □

Based on this mgf of $\underline{x}' A \underline{x}$, the variance of $\underline{x}' A \underline{x}$ can be obtained, which is given by

$$\text{Var}(\underline{x}' A \underline{x}) = 2\text{tr}\left[(A\Sigma)^2\right] + 4\underline{\mu}' A \Sigma A \underline{\mu}.$$

Lemma 3.1. *If A is a $p \times p$ positive definite matrix and B is a $p \times p$ symmetric matrix, then there exists a $\delta > 0$ such that, for $|t| < \delta$, $A + tB$ is positive definite.*

Theorem 3.10. *Let R be a $p \times p$ matrix, T a $q \times q$ matrix, S a $p \times q$ matrix, and U a $q \times p$ matrix. If R and T are nonsingular, then*

$$|R + STU| = |R| \cdot |T| \cdot |T^{-1} + UR^{-1}S|.$$

Corollary. Let S be a $p \times q$ matrix and U a $q \times p$ matrix. Then, $|I + SU| = |I + US|$.

3.3 The Noncentral χ^2 and F Distributions

Definition. A random variable u is said to have a noncentral χ^2 distribution with degrees of freedom n and noncentrality parameter λ if u has the pdf

$$\chi^2(u\colon n,\lambda) = \begin{cases} \displaystyle\sum_{j=0}^{\infty} \left(\frac{e^{-\lambda}\lambda^j}{j!}\right) \left[\frac{u^{(n+2j-2)/2}e^{-u/2}}{\Gamma\left(\frac{n+2j}{2}\right)2^{j+(n/2)}}\right], & \text{for } u > 0, \\ 0, & \text{for } u \le 0, \end{cases}$$

where $\lambda \ge 0$ and $n > 0$.

Note. In the above pdf, $0! = 1$ and $0^0 = 1$.

With $\lambda = 0$, the noncentral χ^2 distribution reduces to the χ^2 or central χ^2 distribution with the pdf

$$\chi^2(u\colon n) = \begin{cases} \frac{u^{(n-2)/2}e^{-u/2}}{\Gamma\left(\frac{n}{2}\right)2^{n/2}}, & \text{for } u > 0, \\ 0 & \text{for } u \le 0. \end{cases}$$

Notation. We use $u \sim \chi^2(n,\lambda)$ to indicate that u is a random variable whose pdf is $\chi^2(u\colon n,\lambda)$, and $u \sim \chi^2(n)$ will be used if the pdf is $\chi^2(u\colon n)$.

Note that

$$\chi^2(u\colon n,\lambda) = \sum_{j=0}^{\infty} \left(\frac{e^{-\lambda}\lambda^j}{j!}\right) \chi^2(u\colon n+2j),$$

and that $e^{-\lambda}\lambda^j/j! = \Pr(v = j)$, where v is a Poisson random variable with parameter λ. Thus, $\chi^2(u\colon n,\lambda)$ is a pdf because $\chi^2(u\colon m)$ is a pdf for each $m > 0$.

For $u \sim \chi^2(n, \lambda)$, we have the mgf of u as

$$M_u(t) = (1-2t)^{-n/2} \exp\left(\frac{2t\lambda}{1-2t}\right), \quad t < \frac{1}{2}.$$

Theorem 3.11. *Let $\underline{x} \sim N(\underline{\mu}, I_n)$. Define w by $w = \underline{x}'\underline{x}$. Then, $w \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\underline{\mu})$.*

Proof. Let $u \sim \chi^2(n, \frac{1}{2}\underline{\mu}'\underline{\mu})$. By Theorem 3.1, it suffices to show that the mgf of w is the same as that of u. From the mgf of a quadratic form in Section 3.2, with $A = I_n$ and $\Sigma = I_n$, we have, for $t < 1/2$,

$$M_w(t) = M_{\underline{x}'\underline{x}}(t) = |I_n - 2tI_n|^{-1/2} \exp\left[t\underline{\mu}'(I_n - 2tI_n)^{-1}\underline{\mu}\right]$$

$$= (1-2t)^{-n/2} \exp\left[\frac{2t\left(\frac{1}{2}\underline{\mu}'\underline{\mu}\right)}{1-2t}\right] = M_u(t).$$

□

Let $u \sim \chi^2(n, \lambda)$ and $\underline{x} \sim N\left([2\lambda/n]^{1/2}\underline{1}_n, I_n\right)$, where $\underline{1}_n$ is the $n \times 1$ vector of 1's. Then,

$$\frac{1}{2}\left([2\lambda/n]^{1/2}\underline{1}_n\right)'\left([2\lambda/n]^{1/2}\underline{1}_n\right) = (\lambda/n)\,\underline{1}_n'\underline{1}_n = \lambda.$$

By Theorem 3.11, $\underline{x}'\underline{x} \sim \chi^2(n, \lambda)$ and so $\underline{x}'\underline{x} \stackrel{d}{=} u$. We then have moments for u.

By Lemma 2.8, we have $E\left(\underline{x}^{'}A\underline{x}\right) = \mathrm{tr}\left(A\Sigma\right) + \underline{\mu}^{'}A\underline{\mu}$. For so $u \sim \chi^2(n, \lambda)$,

$$E(u) = E(\underline{x}^{'}\underline{x}) = \mathrm{tr}\left(I_n\right) + (2\lambda/n)\,\underline{1}_n^{'}\underline{1}_n = n + 2\lambda.$$

By the result for $\mathrm{Var}(\underline{x}^{'}A\underline{x})$ in Section 3.2,

$$\mathrm{Var}(u) = \mathrm{Var}(\underline{x}^{'}\underline{x}) = 2\mathrm{tr}(I_n) + 4\,(2\lambda/n)\,\underline{1}_n^{'}\underline{1}_n = 2(n + 4\lambda).$$

Theorem 3.12. *If $u_1, \ldots, u_m$ are mutually independent random variables and if $u_i \sim \chi^2(n_i, \lambda_i)$ for $i = 1, \ldots, m$, then $u = u_1 + \cdots + u_m \sim \chi^2(n, \lambda)$, where $n = n_1 + \cdots + n_m$ and $\lambda = \lambda_1 + \cdots + \lambda_m$.*

Proof. Let $v \sim \chi^2(n, \lambda)$. By Theorem 3.1, it suffices to show that the mgf of u is the same as that of v. We have

$$M_u(t) = E\left\{\exp\left[t(u_1 + \cdots + u_m)\right]\right\} = \prod_{i=1}^{m} E\left[\exp\left(tu_i\right)\right] = \prod_{i=1}^{m}\left[(1-2t)^{-n_i/2}\exp\left(\frac{2t\lambda_i}{1-2t}\right)\right]$$

$$= (1-2t)^{-(\sum_{i=1}^{m} n_i)/2}\exp\left(\frac{2t\sum_{i=1}^{m}\lambda_i}{1-2t}\right) = (1-2t)^{-n/2}\exp\left(\frac{2t\lambda}{1-2t}\right) = M_v(t).$$

□